

LAB ASSIGNMENT-10

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```
CREATE TABLE EMPLOYEE (  
  ID INT PRIMARY KEY,  
  LAST_NAME VARCHAR(50),  
  FIRST_NAME VARCHAR(50),  
  MIDDLE_NAME VARCHAR(50),  
  FATHER_NAME VARCHAR(50),  
  MOTHER_NAME VARCHAR(50),  
  SEX CHAR(1),  
  HIRE_DATE DATE,  
  ADDRESS VARCHAR(100),  
  CITY VARCHAR(50),  
  STATE VARCHAR(50),  
  ZIP VARCHAR(10),  
  PHONE VARCHAR(15)  
);
```

```
CREATE TABLE SCHOOL (  
  ID INT PRIMARY KEY,  
  NAME VARCHAR(100)  
);
```

-- Mapping table (IMPORTANT for queries)

```
CREATE TABLE EMPLOYEE_SCHOOL (  
  EMP_ID INT,
```

```
SCHOOL_ID INT,  
FOREIGN KEY (EMP_ID) REFERENCES EMPLOYEE(ID),  
FOREIGN KEY (SCHOOL_ID) REFERENCES SCHOOL(ID)  
);
```

```
INSERT INTO SCHOOL VALUES (1, 'Computer Science');  
INSERT INTO SCHOOL VALUES (2, 'Mechanical Engineering');  
INSERT INTO SCHOOL VALUES (3, 'Electrical Engineering');  
INSERT INTO SCHOOL VALUES (4, 'Civil Engineering');  
INSERT INTO SCHOOL VALUES (5, 'Business School');  
INSERT INTO SCHOOL VALUES (6, 'Law School');  
INSERT INTO SCHOOL VALUES (7, 'Medical School');  
INSERT INTO SCHOOL VALUES (8, 'Architecture');  
INSERT INTO SCHOOL VALUES (9, 'Design School');  
INSERT INTO SCHOOL VALUES (10, 'Physics Department');
```

```
INSERT INTO EMPLOYEE VALUES
```

```
(1,'Reddy','Anil','K','Ramesh','Sita','M','2021-05-  
10','Addr1','Hyderabad','Telangana','500001','9999999991'),  
(2,'Sharma','Amit','K','Raj','Meena','M','2018-03-12','Addr2','Delhi','Delhi','110001','9999999992'),  
(3,'Kumar','Arjun','M','Suresh','Latha','M','2022-07-20','Addr3','Chennai','Tamil  
Nadu','600001','9999999993'),  
(4,'Das','Ajay','P','Ravi','Geeta','M','2015-09-15','Addr4','Kolkata','West Bengal','700001','9999999994'),  
(5,'Patel','Anita','L','Mahesh','Rina','F','2019-11-30','Addr5','Ahmedabad','Gujarat','380001','9999999995'),  
(6,'Rao','Kiran','K','Prasad','Lakshmi','M','2012-01-01','Addr6','Vijayawada','Andhra  
Pradesh','520001','9999999996'),  
(7,'Singh','Akash','T','Om','Radha','M','2023-02-10','Addr7','Lucknow','UP','226001','9999999997'),  
(8,'Nair','Anu','P','Gopi','Leela','F','2017-08-25','Addr8','Kochi','Kerala','682001','9999999998'),  
(9,'Iyer','Arvind','S','Hari','Uma','M','2010-06-18','Addr9','Bangalore','Karnataka','560001','9999999999'),  
(10,'Verma','Anjali','R','Dev','Kiran','F','2024-01-01','Addr10','Jaipur','Rajasthan','302001','9999999990');
```

```

INSERT INTO EMPLOYEE_SCHOOL VALUES (1,1);
INSERT INTO EMPLOYEE_SCHOOL VALUES (2,1);
INSERT INTO EMPLOYEE_SCHOOL VALUES (2,2);
INSERT INTO EMPLOYEE_SCHOOL VALUES (3,1);
INSERT INTO EMPLOYEE_SCHOOL VALUES (4,2);
INSERT INTO EMPLOYEE_SCHOOL VALUES (5,3);
INSERT INTO EMPLOYEE_SCHOOL VALUES (6,1);
INSERT INTO EMPLOYEE_SCHOOL VALUES (6,2);
INSERT INTO EMPLOYEE_SCHOOL VALUES (7,4);
INSERT INTO EMPLOYEE_SCHOOL VALUES (8,1);
INSERT INTO EMPLOYEE_SCHOOL VALUES (9,5);
INSERT INTO EMPLOYEE_SCHOOL VALUES (10,1);

```

```

1)

SELECT E.*
FROM EMPLOYEE E
JOIN EMPLOYEE_SCHOOL ES ON E.ID = ES.EMP_ID
JOIN SCHOOL S ON ES.SCHOOL_ID = S.ID
WHERE S.NAME = 'Computer Science'
AND E.ID NOT IN (
SELECT EMP_ID FROM EMPLOYEE_SCHOOL
WHERE SCHOOL_ID != 1
);

```

Output:

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | last_name | first_name | middle_name | father_name | mother_name | sex | hire_date | address | city | state | zip | phone | page |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 10 | Khan | Imran | A | Salim Khan | Farida Khan | M | 2020-02-28 | 5-6-77 Charminar | Hyderabad | Telangana | 500002 | 9701234567 | 11223 |
| 10 | Khan | Imran | A | Salim Khan | Farida Khan | M | 2020-02-28 | 5-6-77 Charminar | Hyderabad | Telangana | 500002 | 9701234567 | 11223 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)

```

```

SELECT EMP_ID

```

```

FROM EMPLOYEE_SCHOOL
WHERE SCHOOL_ID IN (1,2)
GROUP BY EMP_ID
HAVING COUNT(DISTINCT SCHOOL_ID) = 2;

```

```

-> HAVING COUNT(DISTINCT SCHOOL_ID) = 2;
+-----+
| EMP_ID |
+-----+
|      1 |
|      2 |
|      6 |
+-----+
3 rows in set (0.01 sec)

```

```

SELECT S.NAME, COUNT(ES.EMP_ID) AS TOTAL
FROM SCHOOL S
LEFT JOIN EMPLOYEE_SCHOOL ES ON S.ID = ES.SCHOOL_ID
GROUP BY S.NAME;

```

```

+-----+-----+
| NAME          | TOTAL |
+-----+-----+
| Computer Science |    10 |
| Mechanical Engineering |    7 |
| Civil Engineering |    2 |
| Electrical Engineering |    1 |
| Data Science    |    2 |
| AI              |    1 |
| Commerce        |    1 |
| MBA             |    1 |
| Physics         |    0 |
| Chemistry       |    0 |
+-----+-----+

```

```

SELECT S.NAME, COUNT(*) AS TOTAL
FROM EMPLOYEE_SCHOOL ES
JOIN SCHOOL S ON ES.SCHOOL_ID = S.ID
GROUP BY S.NAME
ORDER BY TOTAL DESC
LIMIT 1;

```

```
mysql> SELECT S.NAME, COUNT(*) AS TOTAL
-> FROM EMPLOYEE_SCHOOL ES
-> JOIN SCHOOL S ON ES.SCHOOL_ID = S.ID
-> GROUP BY S.NAME
-> ORDER BY TOTAL DESC
-> LIMIT 1;
```

```
+-----+-----+
| NAME          | TOTAL |
+-----+-----+
| Computer Science |    10 |
+-----+-----+
1 row in set (0.00 sec)
```

```
SELECT DISTINCT E.*
FROM EMPLOYEE E
JOIN EMPLOYEE_SCHOOL ES ON E.ID = ES.EMP_ID
WHERE E.FIRST_NAME LIKE 'A%';
```

id	last_name	first_name	middle_name	father_name	mother_name	sex	hire_date	address	city	state	zip	phone	page
3	Sharma	Anita	P	Mahesh Sharma	Sunita Sharma	F	2022-11-20	23-4-11 Governorpet	Vijayawada	Andhra Pradesh	520002	9988776655	34567
6	Singh	Amit	K	Raj Singh	Sita Singh	M	2019-12-12	56-7-89 Banjara Hills	Hyderabad	Telangana	500034	9090909090	67890

```
SELECT *
FROM EMPLOYEE
WHERE HIRE_DATE >= CURRENT_DATE - INTERVAL 3 YEAR;
```

id	last_name	first_name	middle_name	father_name	mother_name	sex	hire_date	address	city	state	zip	phone	page
1	Kumar	Raj	S	Ramesh Kumar	Lakshmi Devi	M	2024-01-15	12-3-45 MG Road	Vijayawada	Andhra Pradesh	520001	9876543210	12345
2	Reddy	Suresh	K	Narayana Reddy	Padma Reddy	M	2023-06-10	45-6-78 Benz Circle	Vijayawada	Andhra Pradesh	520010	9123456780	23456
8	Yadav	Rohit	T	Mahavir Yadav	Kamla Yadav	M	2023-09-30	67-8-90 Sector 15	Noida	Uttar Pradesh	201301	9812345670	89012

```
SELECT *
FROM EMPLOYEE E1
WHERE E1.CITY IN (
SELECT E2.STATE FROM EMPLOYEE E2
);
```

```
mysql> SELECT *
-> FROM EMPLOYEE E1
-> WHERE E1.CITY IN (
-> SELECT E2.STATE FROM EMPLOYEE E2
-> );
Empty set (0.00 sec)
```

```
CREATE VIEW EMP_VIEW AS
```

```

SELECT CONCAT(E.FIRST_NAME,' ',E.LAST_NAME) AS FULL_NAME,
E.HIRE_DATE,
S.NAME AS SCHOOL_NAME
FROM EMPLOYEE E
JOIN EMPLOYEE_SCHOOL ES ON E.ID = ES.EMP_ID
JOIN SCHOOL S ON ES.SCHOOL_ID = S.ID;

```

```
-> );
```

id	last_name	first_name	middle_name	father_name	mother_name	sex	hire_date	address	city	state	zip	phone	page
5	Patel	Neha	D	Rakesh Patel	Meena Patel	F	2020-08-25	78-5-23 Ring Road	Hyderabad	Telangana	500001	9870012345	56789
7	Das	Priya	M	Suresh Das	Anjali Das	F	2022-04-18	11-2-33 Park Street	Kolkata	West Bengal	700016	9345678901	78901

```

SELECT *
FROM EMPLOYEE
WHERE ID NOT IN (
SELECT EMP_ID FROM EMPLOYEE_SCHOOL
WHERE SCHOOL_ID IN (1,2)
);

```

```

mysql> SELECT E.FIRST_NAME, S.NAME
-> FROM EMPLOYEE E
-> JOIN EMPLOYEE_SCHOOL ES ON E.ID = ES.EMP_ID
-> JOIN SCHOOL S ON ES.SCHOOL_ID = S.ID
-> WHERE TIMESTAMPDIFF(YEAR, E.HIRE_DATE, CURRENT_DATE) > 8;
Empty set (0.00 sec)

```

```

SELECT E.FIRST_NAME, S.NAME
FROM EMPLOYEE E
JOIN EMPLOYEE_SCHOOL ES ON E.ID = ES.EMP_ID
JOIN SCHOOL S ON ES.SCHOOL_ID = S.ID
WHERE TIMESTAMPDIFF(YEAR, E.HIRE_DATE, CURRENT_DATE) > 8;

```

```
DROP VIEW EMP_VIEW;
```

```

mysql> DROP VIEW EMP_VIEW;
Query OK, 0 rows affected (0.01 sec)

```

```
ROLLBACK;
```

```
mysql> ROLLBACK;  
Query OK, 0 rows affected (0.00 sec)
```